

MolStudio

Complete Gap Analysis & Implementation Plan

MolStudio vs PyMOL -- Science + Engineering + Architecture

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MolStudio vs PyMOL: Complete Gap Analysis & Implementation

Science + Engineering + Architecture -- Every Feature, Every Detail

IMPORTANT: This document maps every feature in PyMOL against our MolStudio implementation, identifies gaps, and provides a staged plan to achieve industry-grade parity. Each feature is traced to its scientific foundation with research paper citations.

Executive Summary

Metric	MolStudio (Current)	PyMOL	Gap
Representation Modes	11	14	3 missing
Color Schemes	8	12+	4+ missing
Selection Keywords	~12	45+	33+ missing
Interaction Types Detected	2 (H-bond, Hydrophobic)	6+	4+ missing
Surface Algorithms	3Dmol.js built-in	MSMS / Marching Cubes	Need scientific validation
Electrostatics	[Missing] None	APBS + Vacuum ESP	Critical gap
Measurements	[Missing] None	Distance, Angle, Dihedral	Critical gap
Alignment Methods	1 (align-like)	3 (align, super, cealign)	2 missing
Movie/Animation	[Missing] None	Full movie system	Major gap
Ray Tracing	[Missing] None	Built-in CPU ray tracer	Major gap
Mutagenesis	[Missing] None	Rotamer wizard	Major gap
Scripting/API	[Missing] None	Full Python API	Major gap
Labels	[Missing] None	Full label system	Gap
Electron Density Maps	[Missing] None	isomesh/isosurface	Gap

Stage 1: Foundation Corrections (Week 1-2)

Fix Scientific Accuracy of Existing Features

These are bugs and inaccuracies in code we already have. Must fix before adding new features.

1.1 Hydrogen Bond Detection -- Add Angle Criterion

Aspect	Details
Current Code	Interactions.ts checks only distance: $\leq 2.5 \text{ \AA}$ & $\leq 3.5 \text{ \AA}$ between N/O/S atoms
Problem	No angle check. A backbone N near a carbonyl O at 90 deg is NOT a hydrogen bond -- it's a steric contact. This produces massive false positives.
PyMOL	Uses both distance ($\leq 3.2 \text{ \AA}$ D-A) AND angle ($\geq 120 \text{ deg}$ D-HA)
Science	Baker & Hubbard (1984) DOI: 10.1016/0079-6107(84)90007-590007-5 -- H-bonds require directionality from lone pair alignment

Aspect	Details
Fix	Add D-HA angle calculation. If H position unknown, estimate from D-H vector using VSEPR geometry. Require angle ≥ 120 deg\$.
Implementation	In Interactions.ts, after distance check, compute angle using atan2 of cross/dot products between DH and HA vectors

1.2 Add Missing Interaction Types

Interaction	Distance Cutoff	Angle/Geometry	Science	Status
Salt Bridge	$N+O- \leq 4.0 \text{ \AA}$	Charged N (Lys/Arg) Charged O (Asp/Glu)	Coulomb attraction at biological pH	[Missing] Missing
pi-pi Stacking	Centroid distance $3.3 \text{--} 5.5 \text{ \AA}$	Normal angle < 30 deg\$ (parallel) or 90 deg\$ (T-shaped)	Hunter & Sanders (1990) DOI: 10.1021/ja00170a016	[Missing] Stubbed
Cation-pi	Cation to ring centroid $\leq 6.0 \text{ \AA}$	Cation above ring face	Dougherty (1996) DOI: 10.1126/science.271.5246.163	[Missing] Missing
Halogen Bond	$C-XO/N/S \leq 3.5 \text{ \AA}$	C-XA angle $140 \text{--} 180$ deg\$	sigma-hole electrostatic interaction	[Missing] Missing

Implementation: Create ring detection (DFS cycle search on bonded graph), compute ring centroids and normal vectors, then apply geometric filters.

1.3 Fix Alignment -- Add Iterative Outlier Rejection

Aspect	Details
Current Code	Alignment.ts does single-pass Kabsch on all matched C pairs
Problem	Flexible loops heavily skew RMSD. PyMOL's align iteratively removes pairs with $d > 2\sigma$ from the mean
PyMOL	align: 5 cycles of outlier rejection (default cycles=5, cutoff=2.0 sigma)
Science	Kabsch (1976) DOI: 10.1107/S0567739476001873
Fix	After initial Kabsch, compute per-pair distances. Remove pairs where $d_i > \bar{d} + 2\sigma$. Repeat Kabsch. Iterate 5 times.

1.4 Fix Needleman-Wunsch Scoring Matrix

Aspect	Details
Current Code	Match=1, Mismatch=1, Gap=1 (trivial scoring)
Problem	Doesn't account for amino acid similarity. Leulle substitution should score higher than LeuAsp.
Fix	Replace with BLOSUM62 substitution matrix (standard for structural alignment). Add affine gap penalties (gap open=10, gap extend=1).

Stage 2: Selection Engine Parity (Week 2-3)

Complete the PyMOL Selection Algebra

Our SelectionParser.ts currently supports ~12 keywords. PyMOL has 45+. Here is the complete implementation map:

2.1 Missing Selection Keywords -- Full Inventory

Keyword	What It Selects	PyMOL	MolStudio	Priority	Implementation
name	Atom name (CA, CB, N, O)	[Done]	[Done]	--	Done
resn	Residue name (ALA, GLY)	[Done]	[Done]	--	Done
resi	Residue index (1-10, 50+100)	[Done]	[Done]	--	Done
chain	Chain ID (A, B)	[Done]	[Done]	--	Done
elem	Element symbol (C, N, O)	[Done]	[Done]	--	Done
ss	Secondary structure (H, S, L)	[Done]	[Done]	--	Done
b	B-factor range (b > 40)	[Done]	[Missing]	High	Add numeric comparison in parser
q	Occupancy (q < 1.0)	[Done]	[Missing]	High	Add occupancy to Atom type
alt	Alternate conformer (A, B)	[Done]	[Missing]	Medium	Filter by altLoc field
segi	Segment identifier	[Done]	[Missing]	Low	Add segment parsing
id	Atom serial number	[Done]	[Missing]	High	Match by atom.serial
index	Internal index (0-based)	[Done]	[Missing]	Medium	Use array index
pepseq	1-letter sequence pattern	[Done]	[Missing]	Medium	Sliding window match on translated sequence
model	Object name	[Done]	[Missing]	Low	Need multi-model support
polymer	Protein + nucleic acid	[Done]	[Missing]	High	!isHetero filter
organic	Organic ligands	[Done]	[Missing]	High	isHetero && !isSolvent && hasCarbons
inorganic	Metal ions, inorganic	[Done]	[Missing]	High	isHetero && !hasCarbons && !isSolvent
solvent	Water molecules	[Done]	[Missing]	High	resName in [HOH, WAT, DOD]
hetatm	All HETATM records	[Done]	[Missing]	High	isHetero === true
hydrogens	All hydrogen atoms	[Done]	[Missing]	High	elem === 'H'
metals	Metal ions	[Done]	[Missing]	High	Element in metals set
donors	H-bond donors (N-H, O-H)	[Done]	[Missing]	Medium	N/O with bonded H
acceptors	H-bond acceptors (O, N with lone pair)	[Done]	[Missing]	Medium	N/O not fully protonated
all	Everything	[Done]	[Done]	--	Done
none	Nothing	[Done]	[Missing]	Low	Return empty set
visible	Currently shown atoms	[Done]	[Missing]	Low	Track visibility state
bonded	Atoms with bonds	[Done]	[Missing]	Medium	atom.bonds.length > 0
within X of Y	Distance-based (inside)	[Done]	[Missing]	High	Like around but inverted selection target
around X of Y	Distance-based (exterior)	[Done]	[Done]	--	Done
near_to	Similar to within	[Done]	[Missing]	Medium	Alias for within

Keyword	What It Selects	PyMOL	MolStudio	Priority	Implementation
beyond X of Y	Farther than X from Y	[Done]	[Missing]	Medium	Invert within
gap X of Y	Surface distance	[Done]	[Missing]	Low	VDW-aware distance
byres	Expand to full residue	[Done]	[Done]	--	Done
bychain	Expand to full chain	[Done]	[Missing]	High	Group by chainID
bymolecule	Expand to connected component	[Done]	[Missing]	Medium	BFS on bond graph
byobject	Expand to full object	[Done]	[Missing]	Low	All atoms in model
neighbor	Directly bonded atoms	[Done]	[Missing]	High	1-hop bond graph traversal
extend N	Expand selection by N bonds	[Done]	[Missing]	Medium	N-hop BFS on bond graph
first	First atom in selection	[Done]	[Missing]	Low	selection[0]
last	Last atom in selection	[Done]	[Missing]	Low	selection[N-1]
like	Same name/resn as selection	[Done]	[Missing]	Low	Property matching

Total: 12 implemented / 45 needed = 27% coverage. Must reach 90%+ for industry grade.

Stage 3: Measurement & Label System (Week 3-4)

3.1 Distance Measurement

Aspect	Details
PyMOL	distance dist1, /mol//A/100/CA, /mol//A/200/CA -- draws dashed line with A label
Science	Euclidean distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
MolStudio	[Missing] Not implemented
Implementation	Click-to-measure mode: user clicks two atoms, engine draws a dashed cylinder between them with a floating label showing the distance in Å. Store measurements in Zustand as {atom1, atom2, distance}[] . Render as 3Dmol.js custom shapes.

3.2 Angle Measurement

Aspect	Details
PyMOL	angle ang1, atom1, atom2, atom3 -- shows angle arc with degree label
Science	$\theta = \arccos\left(\frac{\vec{BA} \cdot \vec{BC}}{ \vec{BA} \vec{BC} }\right)$
Implementation	3-atom click mode. Draw two lines meeting at vertex, add arc indicator and degree label.

3.3 Dihedral Measurement

Aspect	Details
PyMOL	dihedral dihed1, a1, a2, a3, a4
Science	$\omega = \text{atan2}()$
Implementation	4-atom click mode. Already have dihedral() function in MolProcessor.ts -- expose to UI.

3.4 Label System

Aspect	Details
PyMOL	label n. CA, resn+resi -- places text at atom positions
MolStudio	[Missing] Not implemented
Implementation	Use 3Dmol.js addLabel() API. Support templates: {resn}, {resi}, {chain}, {name}, {elem}, {b}. Add label controls to ribbon bar.

Stage 4: Representation & Rendering Upgrades (Week 4-6)

4.1 Representation Gap Analysis

Representation	Science	PyMOL	MolStudio	Gap Analysis
Line	Bond connectivity graph	[Done]	[Done]	[Match] Matching
Stick	Cylindrical bond model	[Done]	[Done]	[Match] Matching
Ball-and-Stick	Spheres (0.25x VDW) + sticks	[Done]	[Done]	[Match] Matching
Space-Filling (CPK)	Full VDW spheres (Bondi 1964, DOI: 10.1021/j100785a001)	[Done]	[Done]	[Warning] Verify radii match Bondi values exactly
Cartoon	Richardson diagrams (helices=cylinders, sheets=arrows, loops=tubes). Science: Pauling et al. (1951), DOI: 10.1073/pnas.37.4.205	[Done]	[Done]	[Warning] 3Dmol.js cartoon has limited customization vs PyMOL's cartoon_oval_length, cartoon_flat_sheets
Ribbon	Simplified backbone trace	[Done]	[Missing]	Need ribbon as distinct from cartoon
Surface (SAS)	Lee & Richards (1971), DOI: 10.1016/0022-2836(71)90324-X90324-X). Probe=1.4Å	[Done]	[Done]	[Match] Uses 3Dmol.js SAS
Surface (SES)	Connolly (1983), DOI: 10.1126/science.6879170. Reentrant surface	[Done]	[Done]	[Match] Uses 3Dmol.js SES
Mesh	Wireframe of surface triangulation	[Done]	[Done]	[Match] Matching
Dots	Dot cloud over VDW spheres	[Done]	[Done]	[Match] Custom Fibonacci sphere implementation
Non-bonded Spheres	Small crosses/spheres for unbonded atoms	[Done]	[Done]	[Match] Matching
Unit Cell	Crystallographic cell boundaries from CRYST1 record	[Done]	[Missing]	Need: Parse CRYST1 draw wireframe box using \$a, b, c, \alpha, \beta, \gamma\$
Extent/Bbox	Bounding box of selection	[Done]	[Missing]	Low priority
Putty/Sausage	Tube with radius proportional to B-factor. Science: Debye-Waller factor, $B = 8\pi^2 \langle u^2 \rangle$	[Done]	[Missing]	Need: Map B-factor to tube radius. High scientific value -- instantly shows flexibility.

4.2 Color Scheme Gap Analysis

Color Scheme	Science	PyMOL	MolStudio	Gap
By Element (CPK)	Corey-Pauling-Koltun convention	[Done]	[Done]	[Warning] Verify RGB values match PyMOL's defaults
By Chain	Distinct hue per chain	[Done]	[Done]	[Match]
SS (PyMOL colors)	Helix=Red, Sheet=Yellow, Loop=Green	[Done]	[Done]	[Match]
SS (Jmol colors)	Helix=Magenta, Sheet=Gold	[Done]	[Done]	[Match]
Rainbow (NC)	Sequence position visible spectrum	[Done]	[Done]	[Match]
B-factor	Debye-Waller thermal motion. Blue(rigid)Red(flexible)	[Done]	[Done]	[Warning] Verify gradient matches PyMOL's spectrum b, blue_white_red
By Hydrophobicity	Kyte-Doolittle scale (1982)	[Done]	[Missing]	Need: Map residue hydrophobicity to color ramp
By Charge	Partial charge Red()/Blue(+)	[Done]	[Missing]	Need: Gasteiger charges + color mapping
By Occupancy	Crystallographic occupancy (q)	[Done]	[Missing]	Low priority
Custom Ramps	User-defined color gradients	[Done]	[Missing]	Medium priority
White	Uniform white	[Done]	[Done]	[Match]

Stage 5: Electrostatics & Protein Preparation (Week 6-8)

5.1 Electrostatic Surface Potential

Aspect	Details
PyMOL	util.protein_vacuum_esp (quick) + APBS plugin (rigorous)
Science	Poisson-Boltzmann: $\nabla \cdot [\epsilon(\mathbf{r}) \nabla \phi(\mathbf{r})] = -\kappa^2 \phi(\mathbf{r})$ $-4\pi \rho(\mathbf{r}) / \epsilon(\mathbf{r})$ (Baker et al. 2001, DOI: 10.1073/pnas.181342398)
MolStudio	[Missing] Not implemented
Implementation Plan	Phase A (Quick): Implement vacuum Coulomb ESP -- $\sum \phi(\mathbf{r}) = \sum_i q_i / (\epsilon_r \cdot r)$

5.2 Gasteiger Partial Charge Assignment

Aspect	Details
Science	Gasteiger & Marsili (1980), DOI: 10.1016/0040-4020(80)80168-280168-2)
Current Code	Docking.ts hardcodes charge = 0.000 for all protein atoms
Fix	Implement PEOE iterative algorithm: $\Delta q = (\chi_A - \chi_B) / (\chi_A + \chi_B) \times \text{damping}$. 6 iterations. Electronegativity parameters from Gasteiger's original tables.

5.3 Ramachandran Plot

Aspect	Details
Science	Ramachandran et al. (1963), DOI: 10.1016/S0022-2836(63)80023-6
PyMOL	Available via plugins (e.g., ramachandran_plot)
MolStudio	[Missing] Not implemented
Implementation	Compute ϕ , ψ for every residue using the dihedral() function already in MolProcessor.ts. Render as 2D scatter plot (SVG or Canvas) overlaid on the allowed regions contour map. Color points by SS type. Interactive: click point highlight residue in 3D.

5.4 Protein Preparation Completeness

Feature	PyMOL	MolStudio	Gap
Strip solvent (HOH)	[Done] remove solvent	[Done] stripSolvent()	[Match]
Remove alt conformers	[Done] remove not alt A+"	[Done] filterAltlocs()	[Match]
Add hydrogens	[Done] h_add (valence-based)	[Warning] addHydrogens() (VSEPR heuristic)	[Warning] Our H placement is geometrical only -- no pKa optimization. Good enough for visualization but not for docking prep.
Assign SS	[Done] dss (DSSP-like)	[Done] calculateSecondaryStructure('dssp')	[Match] Matches Kabsch & Sander (1983)
Bond perception	[Done] Automatic	[Done] assignBonds(tolerance)	[Match] Uses covalent radii from Cordero et al. (2008)
pKa estimation	[Warning] Via PROPKA plugin	[Missing]	Gap -- needed for accurate protonation at specific pH
Missing residue modeling	[Missing] Requires MODELLER	[Missing]	Both lack this

Stage 6: Advanced Analysis Tools (Week 8-12)

6.1 Additional Alignment Methods

Method	Algorithm	When to Use	PyMOL	MolStudio
align	Sequence alignment (NW) Kabsch + outlier rejection	Closely related sequences (>30% identity)	[Done]	[Warning] (no outlier rejection)
super	Structure-based dynamic programming Kabsch + outlier rejection	Low sequence similarity	[Done]	[Missing]
cealign	Combinatorial Extension (CE) algorithm	Very distant homologs, fold-level comparison	[Done]	[Missing]

Implementation: super -- use distance matrix alignment instead of NW. cealign -- implement the CE algorithm from Shindyalov & Bourne (1998), DOI: 10.1110/ps.7.11.2480.

6.2 Mutagenesis Wizard

Aspect	Details
PyMOL	Interactive wizard: select residue choose target amino acid browse backbone-dependent rotamers evaluate VDW clashes apply

Aspect	Details
Science	Dunbrack rotamer library (2002), DOI: 10.1002/prot.10286. Backbone-dependent (ϕ, ψ) -binned rotamer frequencies.
MolStudio	[Missing] Not implemented
Implementation	Load Dunbrack rotamer library (JSON format). On residue selection, replace side chain atoms with target amino acid's rotamers. Score each rotamer by VDW clash energy ($\sum \text{LJ}_{\text{repulsive}}$). Display ranked rotamers with clash indicators.

6.3 Electron Density Maps

Aspect	Details
PyMOL	isomesh map_mesh, my_map, 1.5 -- contour at 1.5sigma
Science	Fourier transform of structure factor amplitudes and phases 3D electron density grid
MolStudio	[Missing] Not implemented
Implementation	Parse CCP4/MRC map files marching cubes isosurface at user-specified sigma level render as mesh/surface in 3Dmol.js

Stage 7: Animation, Scripting & Export (Week 12-16)

7.1 Multi-State / Trajectory Support

Aspect	Details
PyMOL	Loads multi-MODEL PDBs, NMR ensembles, MD trajectories as "states"
MolStudio	[Missing] Single state only
Implementation	Parse MODEL/ENDMDL blocks store as state[] array add frame slider in ribbon bar animate with requestAnimationFrame

7.2 Morphing

Aspect	Details
PyMOL	Linear coordinate interpolation between two aligned structures
Implementation	After Kabsch alignment, interpolate: $\mathbf{r}_t = (1-t)\mathbf{r}_A + t \cdot \mathbf{r}_B$ for $t \in [0, 1]$, generate 30 intermediate frames

7.3 Scripting Engine

Aspect	Details
PyMOL	Full Python API (cmd.load(), cmd.color(), etc.) + .pml script files
MolStudio	[Missing] Only selection query console
Implementation	Extend SelectionQueryConsole into a full command interpreter. Support commands: load, fetch, color, show, hide, select, distance, angle, set, get, save, png. Parse commands dispatch to Zustand actions and 3Dmol.js API.

7.4 Session Save/Load (Enhanced)

Aspect	Details
Current	JSON serialization of state (works for basic data)
Upgrade	Support PSE-like session format: include camera matrices, all measurements, labels, custom colors, multiple objects, and named selections

7.5 Export Formats

Format	PyMOL	MolStudio	Priority
PNG screenshot	[Done] png	[Missing]	High -- use canvas.toDataURL()
PDB file	[Done] save	[Warning] toPDB() exists	Verify column widths
PDBQT	[Done] via plugin	[Done] convertToPDBQT()	[Match]
MOL2/SDF	[Done]	[Missing]	Medium
SVG vector	[Done] via draw	[Missing]	Low
VRML/glTF 3D	[Done]	[Missing]	Low

Stage 8: Ray Tracing & Publication Quality (Week 16-20)

8.1 Ambient Occlusion

Aspect	Details
PyMOL	CPU-based ray tracer with ambient occlusion
Science	Casts N rays from each surface point. Ratio of unoccluded rays = ambient light. Darkens cavities and crevices.
Implementation	Use WebGL SSAO (Screen-Space Ambient Occlusion) shader. Cheaper than true ray tracing but visually excellent. 3Dmol.js supports custom shader injection.

8.2 Depth Cue / Fog

Aspect	Details
PyMOL	set depth_cue, 1 -- fades distant objects to background
Implementation	WebGL fog: gl.fog = { near: clipNear, far: clipFar, color: bgColor }

8.3 High-Resolution Export

Aspect	Details
PyMOL	ray 4000, 3000 -- renders at 4K resolution
Implementation	Create off-screen canvas at 4x resolution, render, export as PNG. Alternatively, use Three.js WebGLRenderer.setPixelRatio().

Implementation Priority Matrix

Priority	Stage	Features	Science Papers	Effort
P0	Stage 1	Fix H-bond angles, add salt bridges/pi-stacking, fix alignment outlier rejection	Baker (1984), Hunter (1990), Kabsch (1976)	1-2 weeks
P0	Stage 3	Distance/angle/dihedral measurements, labels	Basic geometry	1 week
P1	Stage 2	Selection engine (add 30+ keywords)	--	2 weeks
P1	Stage 4	Unit cell, putty/sausage representation, hydrophobicity coloring	Bondi (1964), Kyte-Doolittle (1982)	2 weeks
P2	Stage 5	Vacuum ESP, Gasteiger charges, Ramachandran plot	Gasteiger (1980), Ramachandran (1963), Baker (2001)	3 weeks
P2	Stage 6	super/cealign, mutagenesis wizard, electron density	Dunbrack (2002), Shindyalov (1998)	4 weeks
P3	Stage 7	Multi-state, morphing, scripting, export	--	4 weeks
P3	Stage 8	SSAO, fog, high-res export	--	2 weeks

Architecture Decision: Where Science Runs

Computation	Client (Browser/WASM)	Server (Oracle VPS)	Rationale
Bond perception	[Done]		Fast, O(N) with spatial hash
DSSP assignment	[Done]		Our implementation is performant
Selection queries	[Done]		Must be interactive (<50ms)
Measurements	[Done]		Trivial math
Gasteiger charges	[Done]		Iterative but lightweight
Vacuum ESP	[Done]		O(NxM) but feasible for <10K atoms
APBS Poisson-Boltzmann		[Done]	Grid solver needs >4GB RAM
Docking (Webina WASM)	[Done]		Already works client-side
Docking (MolDock C++)		[Done]	Future GPU-accelerated engine
Rotamer scoring	[Done]		Lookup table + VDW clash check
Electron density maps	[Done]		Marching cubes on typed arrays
Ray tracing (SSAO)	[Done]		WebGL shader

Scientific Credibility Checklist

Every distance, angle, and constant in our code must cite its source:

Constant	Value	Source Paper	DOI
H-bond DA distance	≤ 3.2 Å	Baker & Hubbard (1984)	10.1016/0079-6107(84)90007-590007-5)
H-bond D-HA angle	≥ 120 deg	McDonald & Thornton (1994)	10.1006/jmbi.1994.1334
DSSP energy cutoff	0.5 kcal/mol	Kabsch & Sander (1983)	10.1002/bip.360221211
DSSP q1, q2	0.42, 0.20 e	Kabsch & Sander (1983)	10.1002/bip.360221211

Constant	Value	Source Paper	DOI
Water probe radius	1.4 Å	Lee & Richards (1971)	10.1016/0022-2836(71)90324-X90324-X)
VDW radii (C, N, O, S, H)	1.70, 1.55, 1.52, 1.80, 1.20 Å	Bondi (1964)	10.1021/j100785a001
Covalent radii	Per element	Cordero et al. (2008)	10.1039/b801115j
Salt bridge cutoff	≤ 4.0 Å (N+O-)	Barlow & Thornton (1983)	10.1016/0022-2836(83)90148-790148-7)
pi-pi stacking distance	3.3-5.5 Å centroid	Hunter & Sanders (1990)	10.1021/ja00170a016
Cation-pi distance	≤ 6.0 Å	Dougherty (1996)	10.1126/science.271.5246.163
-helix phi, psi	57 deg, 47 deg	Ramachandran et al. (1963)	10.1016/S0022-2836(63)80023-680023-6)
Peptide bond planarity	omega 180 deg (trans)	Pauling et al. (1951)	10.1073/pnas.37.4.205

TIP: The goal: Every scientist who uses MolStudio should be able to look at the "About" or documentation and see that our algorithms cite the same papers that PyMOL, Chimera, and VMD rely on. This is what makes an application industry-grade -- not just features, but scientific trust.